



Fiona Stanley Fremantle Hospitals Group

OFFICIAL

Guideline

Lithium Therapy- Initiation and Continuation

Reference #: FSFHG-HW-GUI-0017

Scope

Site	Service/Department/Unit	Disciplines
Fiona Stanley Hospital Fremantle Hospital	Hospital Wide	Medical, Nursing, Pharmacy

Lithium is a high-risk medication with a narrow therapeutic index. Careful patient selection and monitoring is required to minimise the risk of lithium toxicity.

1. Introduction

Lithium has an established role in the treatment of a number of psychiatric conditions, including prophylaxis of bipolar disorder (BD), acute mania and treatment-resistant depression. It is also occasionally used in the management of schizoaffective disorder and chronic schizophrenia.

Although the exact mechanism of action of lithium is unclear, some of its actions include inhibition of dopamine release, enhancement of serotonin release and inhibiting secondary messenger systems.

Lithium is more effective in preventing manic/hypomanic/mixed episodes than depressive episodes. Predictors of good response include classic mania, family history of a good response, fewer prior mood episodes, complete inter-episode recovery and no neurological impairment. Poorer response is noted in mixed states and rapid cycling. The therapeutic effect occurs gradually and may take up to three weeks.

Lithium is a narrow therapeutic index drug. It is handled by the body in a similar way to sodium; most risk factors for toxicity relate to changes in sodium levels and fluid status.

2. Guideline

2.1. Presentation

- 250mg lithium carbonate immediate release (IR) tablet (Lithicarb®)
- 450mg lithium carbonate slow release (SR) tablet (Quilonum®)
- 127 mg lithium citrate per 1mL oral solution (Auspman®- available only at hospital sites)
- 1mL (127mg) lithium citrate oral solution is equivalent to 50mg lithium carbonate

2.2. Indications

- Treatment of manic episodes in Bipolar Disorder (BD)
- Prophylaxis of manic, hypomanic and/or depressive episodes in BD
- Augmentation of treatment-resistant unipolar depression
- Schizoaffective disorder
- Antipsychotic augmentation in chronic schizophrenia
- Other off-label uses under consultant psychiatrist recommendation

2.3. Contraindications

- Hypersensitivity to lithium or any component of the formulation
- Severe or significant cardiovascular disease

Lithium may cause conduction sinus node dysfunction and conduction blocks resulting in arrhythmias (e.g. sick sinus syndrome) and other electrocardiogram (ECG) changes such as QTc prolongation.

- Severe or significant renal disease – risk of lithium toxicity is increased in severe or significant renal impairment associated with medical conditions or concomitant nephrotoxic medications
- Severe sodium depletion and conditions associated with hyponatraemia e.g. Addison's disease, dehydration, severely debilitated patients and patients on low sodium diets
- Frank hypothyroidism

2.4. Precautions

- Pregnancy: Category D. May increase the risk of congenital heart defects and the decision to continue should be made between the Consultant and the patient. Consult a Pharmacist for further information. Lithium in pregnancy should only be used after a careful risk-benefit analysis.
- Lactation: Consult a Pharmacist for information as evidence is constantly changing.

- Older adults: excretion may be reduced so lower dosages, slower dose titration and more frequent monitoring may be required to avoid toxicity.
- Renal impairment: Lithium is primarily excreted by the kidneys, so caution is required. Doses should be reduced, monitoring performed more frequently, and lithium levels aimed at the lower end of the therapeutic range.
- Hyponatremia: depleted sodium levels increase serum lithium levels and therefore increase toxicity.
- Hypothyroidism: Lithium can cause thyroid dysfunction therefore use with caution in patients with thyroid disease and consider increased thyroid monitoring.
- Concomitant serotonergic medication: Lithium can contribute to serotonin toxicity when used in combination with other serotonergic medications.
- Psoriasis: may worsen with lithium
- Electroconvulsive therapy (ECT): refer to section 2.13. Peri-operative and ECT Management of Lithium
- Obesity: Lithium clearance is significantly greater for obese patients than for non-obese. Evidence suggests obese patients may therefore need a higher maintenance dose, but dosing must be serum-level guided. Use lean bodyweight as a guide for initial dosing, and then titrate to target serum-level with close monitoring

2.5. Pre-initiation work-up and ongoing clinical monitoring of lithium

- Patients commenced on lithium must undergo appropriate baseline and ongoing clinical and laboratory investigations aimed at optimising therapy and minimising risk of toxicity.
- Obtain personal and family medical history including thyroid function, previous heart disease, renal disease and best possible medication history.
- Prior to initiating lithium, baseline investigations outlined below should be performed. The below table also outlines ongoing monitoring requirements post-initiation.

Pre-initiation work-up	Monitoring and comments
Full blood picture Thyroid function tests	Repeat after 3 months and then 6 monthly Monitor for clinical signs and symptoms of thyroid dysfunction Perform on admission if already taking lithium
Creatinine Urea and electrolytes, including	Repeat every 3-6 months Repeat renal function tests more frequently

calcium, phosphate, magnesium	if there is renal instability Perform on admission if already taking lithium
Fasting blood glucose levels	Repeat after 6 months
Weight, body mass index (BMI) and waist circumference	Repeat after 3 months
Calcium (including ionised calcium) Phosphate Parathyroid hormone	Repeat every 1-2 years
ECG (required for patients: >65yo, with cardiac disease or risk factors for cardiac disease, and/or taking other medications which may prolong QT interval)	Repeat annually if normal as per mental health requirement or when clinically indicated
Beta HCG	Ensure adequate contraception, lithium should be avoided in pregnancy

- For more information on lithium monitoring during initiation and maintenance therapy, refer to section 2.8 Therapeutic Drug Monitoring

2.6. Drug Interactions

As lithium is a narrow therapeutic index drug, pharmacokinetic interactions with other drugs can precipitate lithium toxicity.

The following drug interactions are limited to those which occur most commonly in practice. Drug interactions with lithium are extensive so multiple references should be consulted beyond this document, including the Product Information (available via [AusDI](#)), [Australian Medicines Handbook](#) and Clinical Pharmacist advice.

Concomitant use of lithium with interacting medications requires closer monitoring of serum lithium concentration, renal function and clinical effects for extended periods. Consider use of an alternative agent where possible, and/or adjust the dose of lithium when necessary.

- **Angiotensin Converting Enzyme (ACE) inhibitors and Angiotensin II Receptor antagonists:** can reduce lithium clearance and therefore increase lithium levels and the risk of toxicity. May take several weeks to develop.
- **Diuretics:** (e.g. thiazides and loop diuretics) can reduce lithium clearance and therefore increase lithium levels and the risk of toxicity. Increase in lithium levels usually occur within 10 days of thiazide initiation and the magnitude is unpredictable. Increases in levels with loop diuretics is greatest in the first month after initiation.

- **NSAIDs:** (e.g. ibuprofen) can reduce lithium clearance and therefore increase lithium levels and the risk of toxicity. Avoid the combination where possible. Low dose aspirin is safe to use.
- **Serotonergic drugs:** Lithium can contribute to serotonin toxicity, therefore patients who are prescribed combinations of serotonergic drugs should be closely monitored
- **Anti-psychotics:** rapid dose increases of lithium and antipsychotics together may cause neurotoxicity.
- **Urinary alkaliniser's:** (e.g. sodium bicarbonate and potassium citrate (found in cystitis-relief sachets and some antacids) can theoretically increase lithium clearance and thereby reduce efficacy. However, it is likely that high doses would need to be consumed to have a significant effect.

2.7. Dosage (as lithium carbonate)

- Doses should be individualised depending on indication and patient risk factors i.e. weight, comorbidities (e.g. renal impairment) and concomitant medications, and titrated based on serum lithium levels (refer to section 1.9. Therapeutic Drug Monitoring), tolerability and clinical response.
- The total daily dose of lithium should not exceed 2500mg.
- When switching from the IR (250mg tablet) to SR formulation (450mg tablet), give the total daily dose that is closest to that which is currently prescribed.
- Single night-time dosing with slow release (SR) tablets may improve compliance and facilitate morning blood tests.
- Twice daily dosing should be spaced by 12 hours
- It is recommended that lithium be taken with food.
- Reduce doses in the elderly and in patients with renal impairment.
- It may take up to three weeks to observe a therapeutic effect following initiation.
- Acute Mania:
 - IR product: 750 to 1000mg daily in 2 or 3 divided doses or as a single dose at night.
 - Increase daily dose in 250 to 500mg increments based on serum lithium levels (refer to section 1.9 Therapeutic Drug monitoring)
 - A daily dose of 1000 to 2500mg is usually required, except in elderly who may require a third to half this dose due to poor tolerability
 - SR product: 450 to 900mg daily in 2 divided doses or as a single dose at night (usual is 900mg daily).

- Increase daily dose in 225 to 450mg increments based on serum lithium levels (refer to section 1.9 Therapeutic Drug monitoring)
- A daily dose of 900 to 2250mg is usually required, except in elderly who may require a third to half this dose due to poor tolerability
- Prophylaxis of bipolar disorder:
 - IR product: 250 to 1000mg daily in 2 or 3 divided doses or as a single dose at night
 - Increase daily dose in 250 to 500mg increments based on serum lithium levels (refer to section 1.9 Therapeutic Drug monitoring)
 - SR product: 450 to 675mg daily in 2 divided doses or as a single dose at night
 - Increase daily dose in 225 to 450mg increments based on serum lithium levels (refer to section 1.9 Therapeutic Drug monitoring)

2.8. Therapeutic Drug Monitoring

- Blood taken for serum lithium levels should be taken 12 hours after the last dose. Doses that would be given prior to blood being drawn should be withheld until after blood has been taken.
- Steady state concentration is achieved after 5-7 days. This may be longer (between 7-10 days) in the elderly or those with renal impairment.
- Levels should be assessed:
 - On admission to hospital for patients currently on lithium therapy
 - 5-7 days after initiating treatment
 - Once weekly for the first four weeks if initiated as an inpatient
 - 5-7 days after each dose change
 - After the initiation, change or withdrawal of interacting medications
 - Every three months after a stable therapeutic concentration has been achieved.
- More frequent monitoring is required in certain situations such as; illness (any infection, diarrhoea or vomiting), acute mania or depression, changes in diet, pregnancy, concurrent use of interacting medication, changes in mental state or when there are signs of toxicity.
- Target serum lithium ranges are as follows:
 - **Acute mania:** 0.8-1.2 mmol/L (twice daily or three times daily dosing). Once remission occurs, gradually decrease to a maintenance dose that provides a lithium level in the BD prophylaxis range.
 - **Prophylaxis of bipolar disorder:** 0.4-1.0 mmol/L (usually 0.6-0.8mmol/L once stabilised). Older adults who are at a greater risk of cognitive and

renal impairment may benefit from lower serum target levels (0.4-0.7 mmol/L)

- **Augmentation of antidepressant therapy:** 0.4-0.6 mmol/L (optimal plasma level range unclear, if no response within 1 month, psychiatrist to review appropriateness)
- Target concentrations in the upper end of the therapeutic range should be aimed for if the patient is prescribed a single nighttime dose of lithium, because a once daily dose produces a 10-25% higher 12-hour serum concentration:
 - For example, in patients prescribed lithium for prophylaxis of bipolar disorder, the target range for a single nightly dose may be 0.7-1 mmol/L compared to more frequent daily dosing (0.6-0.8 mmol/L).
- Reduce doses and monitor closely in patients with renal impairment.

2.9. Adverse effects

- These are generally dose-related and can potentially be prevented by ensuring lithium levels remain in the therapeutic range. Dose reduction may assist with resolution.

Very common (>10%)	Common (>1-10%)	Infrequent (0.1-1%)	Rare (< 0.1%)
Tremor Metallic taste Nausea Diarrhoea Epigastric discomfort Polydipsia Polyuria	Weight gain Fluid retention Fatigue Headache Vertigo Acne Psoriasis Leucocytosis Hyperparathyroidism Nephrotoxicity Hypothyroidism (usually asymptomatic) Benign T wave changes on ECG	Memory impairment Hair loss	Renal impairment Blurred vision Confusion Arrhythmias Hyperthyroidism

- Nephrotoxic effects:
 - Lithium can cause several nephrotoxic effects including nephrogenic diabetes insipidus, reduced glomerular filtration rate progressing to end-stage renal disease, tubular atrophy, focal interstitial nephropathy and nephrotic syndrome (rare). Some renal insult may be reversible upon cessation of lithium.
 - The long-term use of lithium is associated with an increased risk of chronic kidney disease (insidious onset in 15-20% of patients). Risk factors for

kidney injury include duration of lithium exposure, cumulative dose and advanced age.

2.10. Lithium Toxicity

- Signs and symptoms of toxicity, which may be due to acute (overdose) or chronic causes, varies dependant on serum lithium levels. A toxicologist should be consulted if poisoning is suspected.
- **Signs and symptoms of mild to moderate toxicity**: drowsiness, blurred vision, mild tremors, increasing diarrhoea, nausea, vomiting, anorexia, muscle weakness, apathy, ataxia and flu-like illness.
- **Signs and symptoms of severe toxicity**: increased muscle tone, hyperreflexia, myoclonic jerks, coarse tremor, dysarthria, disorientation, psychosis, seizures, coma, QT-interval prolongation, and death.
- Signs and symptoms of toxicity are common at serum lithium levels **above 1.5mmol/L**, however, may occur at a lower level of **1.2mmol/L** in older patients or those with neurological damage/conditions (e.g. epilepsy)
- Causes of lithium toxicity:
 - Drug interactions altering renal clearance of lithium (refer to 1.7)
 - Significantly reduced salt intake
 - Dehydration
 - Reduced fluid intake
 - Fluid loss from vomiting, diarrhoea, high temperature or excessive sweating
 - Renal impairment
 - Deliberate or inadvertent overdose
- Chronic poisoning: Also known as the 'Syndrome of Irreversible Lithium Effectuated Neurotoxicity' or SILENT, this is associated with significant morbidity and mortality. It has an insidious onset and often requires treatment with intravenous fluids and possibly dialysis. Clinical recovery can take days to weeks and is significantly delayed compared with the decrease in serum lithium concentrations. A Toxicologist should be consulted if poisoning is suspected.

If a patient experiences any symptoms of toxicity (mild, moderate or severe), lithium should be withheld, a level taken, and clinical review undertaken to assess whether to dose adjust or cease.

2.11. Patient Counselling

Patient counselling must include possible contributors and symptoms of lithium toxicity and when to seek urgent medical review for possible dose adjustment or temporary cessation of lithium.

Staff should ensure that patients/carers have achieved a satisfactory level of understanding on the information provided.

- Upon commencement of lithium, all patients (including their carer(s) where appropriate) must be provided with detailed written and verbal education on lithium, with a record of this made in the patient health records. Material should be provided in a language and format that is understood by the consumer when available.
 - Advise patients of the likely adverse effects, including signs and symptoms of toxicity, and contributing factors.
 - Advise of the requirement for regular blood tests for routine clinical and serum lithium monitoring through their GP and/or community mental health team.
 - Discuss the importance of maintaining a normal diet without excessive changes in salt, alcohol or fluid intake.
 - Advise to take tablets with food to minimise stomach upset and swallow SR tablets whole.
 - Avoid over-the-counter anti-inflammatories, indigestion medications and urinary alkalisers unless in consultation with a Doctor or Pharmacist.
 - Advise of the importance of taking medication regularly and to avoid abrupt cessation to prevent relapse.
- Patient must be given a WA Department of Health [CHOICE Medication Leaflet](#) to support the verbal counselling.

2.12. Cessation of therapy

- Abrupt cessation of lithium can lead to relapse of mania or depression within a few weeks to months in many patients. Lithium should therefore be withdrawn slowly, preferably over 3 months, under the supervision of a mental health medical team if it is to be ceased.

2.13. Peri-Operative and ECT Management of Lithium

- Lithium can prolong neuromuscular blockade. Recommendations vary from withholding 7 days before surgery to continuing throughout the perioperative period.
- It is recommended that lithium should not be withheld but consider monitoring serum levels before and after surgery, especially if renal function is compromised. However, the ultimate decision should be made between the

managing Anaesthetist, Psychiatrist and Surgeon depending on patient risk factors.

- Concurrent ECT and lithium may increase the risk of developing delirium, prolonged seizures, neurotoxicity and post-ECT confusion, especially in older patients. If lithium is to be continued during the ECT course, reduce lithium to the lower end of the therapeutic range prior to ECT, and withhold the lithium dose the night before each ECT session. The treating Psychiatrist, with Clinical Pharmacist assistance, should thoroughly review lithium level prior to and throughout the course of ECT treatment, as well as post-ECT cognition and seizures.
- It is important to closely monitor the patient's fluid status, as fasting and changes in the patient's fluid status may alter serum lithium concentration.

3. Compliance/performance monitoring

- Compliance with this document will be monitored via relevant staff designations in accordance with standard clinical incident review requirements.
- Adherence to these guidelines may be monitored by clinical audits performed by the Medication Safety Pharmacist.

4. Related policy documents

- FSFHG:
 - High Risk Medications
 - Preoperative and preintervention medication management
 - Electroconvulsive Therapy

5. Related standards

- NSQHS standards:
 - Partnering with consumers
 - Medication Safety
 - Communicating for safety
 - Recognising and responding to acute deterioration

6. Bibliography

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7. Authorisation

EXECUTIVE SPONSOR: Medical Director, Service 2					
Version	Date Issued	Compiled/Revised By	Committee/Consumer Group Consulted	Endorsed By	Revision due
1	03/2015	FSH Senior Clinical Pharmacist	Hospital Executive Committee	Policy or Executive Committee	03/2018
2	06/2021		DTC	FSFHG Policy Committee	06/2025
3	11/2025	FSH WCMHR Team Lead	DTC NMPGC	FSFHG Policy Committee	11/2029